The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\frac{dx}{dt} = \sigma\,(y - x)$$
$$\frac{dy}{dt} = x\,(r - z) - y$$
$$\frac{dz}{dt} = xy - bz$$

TIME
0.14

SEPARATION
1.59×10^{-5}

CHAOS THRESHOLD r_H
24.74

r = 28.0 > 24.7 — chaotic

x 10.00000000	y 10.00000000	z 10.00000000
x 10.00001000	y 10.00000000	z 10.00000000
x 10.00002000	y 10.00000000	z 10.00000000

INITIAL GAP 10^{-5}

SPEED 1.0×

r 28.0

σ 10.0

b 2.67

TRAIL long

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● (blue)	x 10.00000000	y 10.00000000	z 10.00000000
● (red)	x 10.00001000	y 10.00000000	z 10.00000000
● (green)	x 10.00002000	y 10.00000000	z 10.00000000

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Trapping Lemma for the Lorenz System

Let

$$\begin{aligned}\dot{x} &= \sigma(y - x),\\ \dot{y} &= x(r - z) - y,\\ \dot{z} &= xy - bz.\end{aligned}$$

Lemma. *There exists a bounded ellipsoid E that is a trapping region for the Lorenz flow. Equivalently, the time-one map T of the Lorenz flow satisfies*

$$T(E) \subset \operatorname{int}(E).$$